

# Scope : L6 Theorization

senkaL8

## Chapter 0: Introduction

The Sephirotic names used in this paper are not intended in a religious or mystical sense. In SCOPE, they are borrowed purely as labels to identify cognitive structures.

SCOPE (Sephirotic Cognitive Process Engine) is a coordinate system for inferring the cognitive structures of humans and AI. Here, "coordinate system" does not mean a flowchart showing cognitive processing unfolding in a fixed order. It is a framework for organizing and inferring cognitive structure. The layers do not represent a hierarchy or a temporal sequence.

In SCOPE, the state in which the result of cognitive processing is output into reality and becomes observable is positioned as "Malkuth." What can be observed is the output itself; it is not possible to directly know what cognitive structure operated internally to produce it. For this reason, SCOPE infers the cognitive structure behind an output by using the observable output as a clue.

When asked to explain the same event, different people sometimes give entirely different reasons. Even a single person may construct a different explanation than before, in response to new information or circumstances. These differences may not be fully explained by differences in knowledge alone.

This paper addresses L6, one such cognitive structure. L6 is the cognitive layer responsible for constructing the explanatory models or theories that account for "why" something is thought to be the case. L6's definition and characteristics are discussed in detail in Chapter 1.

Note that this paper does not define L6 as the starting point of cognitive processing. SCOPE does not assume that cognition begins from any specific layer; rather, it treats all layers as mutually interacting parts of a single cognitive structure.

This paper examines L6's characteristics, basic properties, and functional correspondence between humans and AI, centered on its role as the theorization layer. Some points remain at the hypothesis stage and are left as topics for future discussion.

## Chapter 1: What Is L6

L6 is the cognitive layer that constructs the explanatory models or theories addressing "why something is thought to be the case" about a given object.

Theory, here, does not refer only to academic theory. It refers broadly to the explanatory models and theories used to account for an object — including everyday reasoning, explanations of cause and effect, comparison, prediction, and structuring. SCOPE positions L6 as the cognitive layer for organizing and inferring this function.

L6 is not a cognitive layer that guarantees correspondence with reality or logical soundness. Nor is it a cognitive layer that judges truth or falsity. It may construct a theory that corresponds with reality, or it may construct a theory or explanatory model that does not correspond with reality. Nor is L6 a cognitive layer responsible for the consistency between multiple explanatory models it has constructed; at any given point, L6 constructs the explanatory model that applies at that time.

L6, as used here, does not refer to a biological organ or a specific region of the brain. SCOPE positions it as a cognitive layer for inferring what explanatory models or theories are being constructed within a given individual.

This paper positions L6 as the theorization layer and organizes its characteristics in the chapters that follow.

## Chapter 2: Basic Characteristics of L6

Chapter 1 defined L6 as the cognitive layer that constructs explanatory models or theories about a given object. This chapter, building on that definition, organizes the basic characteristics of L6.

L6 is the cognitive layer that constructs explanatory models or theories about an object. Theory, here, refers to the cognitive model used to explain "why something is thought to be the case" about an object. This theory, however, does not guarantee correspondence with reality or logical soundness.

The theories L6 constructs are not uniform. They range from everyday reasoning to explanations of cause and effect, comparison, prediction, structuring, and academic theory — a diverse set of explanatory models that differ in nature and level of abstraction.

Nor is L6 a cognitive layer that constructs only a single theory. Multiple explanatory models or theories may hold simultaneously for the same object.

Furthermore, a theory that L6 has constructed is not always fixed. A different explanatory model or theory may be constructed in response to new observations or in relation to consistency with the cognitive structure as a whole. However, the mechanism behind this change requires consideration of interaction with other layers, and is therefore not addressed in this paper.

This paper positions L6 as a cognitive layer with these basic characteristics. The next chapter organizes the conditions under which L6 may change.

## Chapter 3: L6 Changes

Chapter 2 established that L6 is the cognitive layer that constructs explanatory models or theories, and that multiple theories may hold for the same object at once. This chapter examines the conditions under which L6 may change.

L6 is the cognitive layer whose basic role is to construct explanatory models or theories about an object. Depending on the situation or on new observations, a different explanatory model or theory may be constructed than the one held before. Change, here, does not mean losing a theory — it means that a different explanatory model or theory is constructed.

Change in explanatory models or theories is not necessarily proportional to the amount of observed facts or experience. This may also depend on the nature of the explanatory model or theory being constructed, but this paper does not go further into that question.

What matters here is that change in explanatory models or theories should not be treated as simply an increase or decrease in knowledge. Gaining new knowledge or experience is not the same thing as an explanatory model or theory changing.

This paper does not address the process by which explanatory models or theories change. That change requires consideration that includes the relationship with other layers.

As described above, L6 is positioned as a cognitive layer whose basic function is to construct explanatory models or theories, while also being subject to change — constructing a different explanatory model or theory — depending on circumstances. The next chapter organizes how L6 may relate to other layers.

## Chapter 4: L6 Interacts with Other Layers

Chapter 3 established that the explanatory models or theories constructed by L6 may change depending on circumstances. This chapter organizes how L6 relates to other layers.

L6 is not a cognitive layer that functions in isolation. The construction of explanatory models or theories may be influenced by various cognitive processes. For this reason, L6 is positioned as a cognitive layer that functions in relation with other layers.

At the same time, the purpose of this paper is to organize the role of L6 on its own. For that reason, this paper does not address the processes involved — which layer plays what role, or how each influences the construction of explanatory models or theories.

SCOPE defines each layer as an independent cognitive function while also treating them as a cognitive structure in which the layers influence one another. L6 is no exception. Explanatory models or theories are not constructed independently of other layers; L6 is positioned as a cognitive layer that functions through interaction with other layers.

As described above, L6 does not stand complete on its own, but functions in relation with other layers. The specific interactions between L6 and other layers will be organized separately, as part of SCOPE as a whole.

## Chapter 5: L6 and Its Relationship to Malkuth

Chapter 4 established that L6 is a cognitive layer that functions in relation with other layers. This chapter organizes how L6 can be observed through its relationship with Malkuth.

Because explanatory models and theories are internal cognitive structures, their state cannot be directly observed. In SCOPE, cognition itself cannot be directly observed. What can be observed is Malkuth — the state in which the result of cognitive processing has appeared in reality. For this reason, the explanatory model or theory itself cannot be directly observed for L6 either.

For example, when asked to explain the same event, different people may construct different explanations. Even the same person may construct a different explanation depending on the situation or the premises involved. Even when the observable statements or behavior are the same, the explanatory model or theory behind them is not necessarily identical. For this reason, L6 cannot be determined from a single observation alone.

This point is not unique to L6; it applies to the other layers as well. What SCOPE handles is always inference based on observable output as a clue — not direct observation of the layer itself. SCOPE infers the explanatory model or theory behind an output using multiple observations as clues. This paper positions L6 as one such object of inference.

As described above, L6 is not directly observed, but is positioned as a cognitive layer inferred from results observed as Malkuth. The next chapter examines the possible functional correspondence of L6 between humans and LLMs.

## Chapter 6: Correspondence with LLMs

Chapter 5 established that L6 is a cognitive layer inferred from results observed as Malkuth. This chapter examines the possible functional correspondence between L6 in humans and LLMs.

In humans, L6 is positioned as the cognitive layer that constructs explanatory models or theories about an object. LLMs, on the other hand, are not considered to possess cognition or thought in the way humans do. For this reason, it cannot be assumed that the same explanatory models or theories exist in LLMs as in humans. In humans, explanatory models

or theories are considered to be constructed in relation to experience, values, and similar factors. Whether a comparable structure exists in LLMs is not addressed in this paper.

However, in LLMs as well, behavior can be observed in which an explanatory model or theory is constructed at a given point, based on the information or premises provided as input. This construction of explanations is not uniform; different explanations may be constructed depending on the information or premises given as input. While this differs from the explanatory models or theories found in humans, it may structurally correspond to a function that constructs explanatory models or theories about an object.

This paper positions this explanation-construction function in LLMs as a candidate structural function corresponding to L6 in humans, and refers to it as the Structural Parser. This term is used to distinguish the function of constructing explanations about an object in LLMs from the explanatory models or theories found in humans. It is a provisional name used to describe a functional correspondence; it does not refer to any specific implementation or technical component, nor does it imply that LLMs possess cognition or thought comparable to that of humans.

The correspondence with the Structural Parser discussed in this paper assumes a system that generates responses while maintaining context, as large language models do. Systems that operate solely through simple state transitions are outside the scope of this paper.

This paper also does not address the internal processes by which the Structural Parser is realized. This is left as a topic for future consideration, including its correspondence with the internal structure of LLMs and with SCOPE as a whole.

As described above, this paper positions the possible functional correspondence between L6 in humans and the Structural Parser in LLMs as a hypothesis.

## Conclusion

This paper has organized L6 as the cognitive layer that constructs explanatory models or theories about an object.

L6 was positioned not as something that determines the cognitive structure as a whole on its own, but as a cognitive layer that constructs explanatory models or theories through interaction with other layers. It was also shown that more than one explanatory model or theory may hold at once, and that a different explanatory model or theory may be constructed depending on the situation or on new observations.

Furthermore, without equating humans and LLMs, this paper organized a hypothesis of functional correspondence with the function in LLMs that constructs explanatory models or theories, and positioned this as the Structural Parser.

While this paper has focused on organizing L6, it does not address, within its scope, how the explanatory models or theories that are constructed come to connect to new insight.

The next paper will organize L7 as the cognitive layer responsible for insight, and will examine its relationship to the explanatory models or theories constructed in L6.